INCH-POUND

MIL-PRF-26392E

13 November 2007

SUPERSEDING

MIL-PRF-26392D (USAF)

26 December 1996

PERFORMANCE SPECIFICATION

DUMMY CONVERTERS, LIQUID OXYGEN INDICATOR SYSTEM, 10-LITER CRU-23/A AND 25-LITER CRU-24/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of dummy converters (electrical capacitors) having a capacitance equivalent to an empty 10-liter and 25-liter liquid oxygen converter.

1.2 Classification.

Type	Designation
I	CRU-23/A
II	CRU-24/A

1.3 Part or Identifying Number (PIN). The PIN is as follows:

PIN	Type	Designation	Liters
M26392-1	I	CRU-23/A	10
M26392-2	II	CRU-24/A	25

Comments, suggestions, or questions on this document should be addressed to: Oklahoma City Air Logistics Center/ENEP, 3001 Staff Drive, Suite 2AG-68A, Tinker AFB, OK 73145-3036 or e-mailed to ocalc.enrs.sma@tinker.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil/.

AMSC N/A FSC 1660

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. These lists do not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification For

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. All materials shall be suitably treated to resist corrosion due to electrolytic decomposition or other conditions that may be encountered during operational use or storage.

3.4 Interface.

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3.4.1 Dimensions. The dummy converter case, mounting plate, and electrical connectors shall conform to the dimensions in figure 1.

3.4.2 Electrical connectors. The high impedance sensing element lead shall be terminated in a connector conforming to MIL-DTL-25516, B polarity. The low impedance lead shall be terminated in a connector conforming to MIL-DTL-25516, E polarity. The connectors shall provide for waterproof connections and have dust caps to prevent the entrance of foreign particles. The dust caps shall be tethered to prevent movement or loss when not installed on the connectors.

3.5 Performance.

3.5.1 Capacitor. The Type I dummy converter capacitor shall have an electrical capacitance of 123.5 ± 1.0 picofarads (pF), and the Type II shall have an electrical capacitance of 303.5 ± 2.5 pF. Capacitance shall remain constant at temperatures between -65° and 260°F (see 4.6.2, 4.6.3, and 4.6.4). The rated voltage of the capacitors shall be 110V.

3.5.2 Case leakage. The case shall provide a moisture-tight seal (see 4.6.5).

3.5.3 Reliability. When specified (see 6.2), the dummy converter shall be tested to demonstrate a mean time between failures (MTBF) not less than 300 hours. A MTBF not less than 200 hours may also be acceptable if the dummy converter can be tested to demonstrate a 99.33% reliability of completing a 2 hour mission (see 4.6.6).

3.5.4 Maintainability. The total corrective maintenance time for field level expected failures shall be not greater than 1 hour (see 4.6.7).

3.5.5 Weight. The weight of the complete dummy converter shall be not greater than 0.25 pounds.

3.5.6 Interchangeability. All parts having the same manufacturer’s part number shall be functionally and dimensionally interchangeable.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one dummy converter and consist of all examinations and tests specified (see 4.6).

4.3 Conformance inspection. Conformance inspections shall include the individual tests

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in 4.3.1 and the sampling tests in 4.3.2.

4.3.1 Individual tests. Each dummy converter shall be subjected to the following examination and tests:

a. Examination of product (see 4.6.1).

b. Capacitance at room temperature (see 4.6.2).

4.3.2 Sampling tests. One percent of each order (a minimum of one dummy converter), shall be subjected to the following tests:

a. Capacitance at low temperature (see 4.6.3).

b. Capacitance at high temperature (see 4.6.4).

c. Case leakage (see 4.6.5).

4.4 Inspection conditions. Unless otherwise specified, all tests shall be performed at ambient temperature and pressure, and not greater than a relative humidity of 80 percent.

4.5 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

TABLE I: Requirements cross-reference matrix

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.2	4.6.5
3.3	4.6.1	3.5.3	4.6.6
3.4.1	4.6.1	3.5.4	4.6.7
3.4.2	4.6.1	3.5.5	4.6.1
3.5.1	4.6.2, 4.6.3, 4.6.4	3.5.6	4.6.1

4.6 Tests.

4.6.1 Examination of product. Each dummy converter shall be examined to determine conformance to materials, weight, and dimensions.

4.6.2 Capacitance at room temperature. Using a capacitance measuring instrument, the capacitance at room temperature shall be 123.5 ± 1.0 pF for Type I dummy converters and 303.5 ± 2.5 pF for Type II dummy converters.

4.6.3 Capacitance of low temperature. The dummy converter shall be cold soaked at a temperature not greater than -65°F for no less than 8 hours. While still at this temperature, the capacitance shall be 123.5 ± 1.0 pF for Type I dummy converters and 303.5 ± 2.5 pF for Type

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II dummy converters.

4.6.4 Capacitance at high temperature. The dummy converter shall be maintained at a temperature not less than 260°F for no less than 8 hours. While still at this temperature, the capacitance shall be measured and shall be 123.5 ± 1.0 pF for Type I dummy converters and 303.5 ± 2.5 pF for Type II dummy converters.

4.6.5 Case leakage. The dummy converter, in normal operating configuration, shall be heated to 113° ± 5°F and maintained at this temperature for no less than 2 hours. The dummy converter then shall be immersed in water at 64°± 9°F to a depth of 36 ± 5 inches for 2 hours. Any evidence of water penetration into the dummy converter shall be a cause for rejection.

4.6.6 Reliability. This test shall be conducted with sufficient duration to obtain a 90% confidence that a MTBF of 300 hours has been achieved. This test may also be used to demonstrate a MTBF of 200 hours and a 99.33% probability of completing a 2-hour mission. The dummy converter shall be stabilized at a temperature not greater than -65°F, then raised to a temperature not less than 260°F and held for no less than 2 hours. After 2 hours, the temperature shall be reduced to not greater than -65°F. The cycling time between temperature extremes shall be not greater than 40 minutes. The dummy converter shall be subjected to not less than four of these temperature cycles every 24 hours. The dummy converter shall be subjected to vibration at a frequency between 20 and 60 Hz and an applied acceleration of ± 2g for not less than 10 minutes during each hour of this test. The rated voltage within ± 2% shall be applied continuously during this test. The dummy converter shall be checked at least once every 24 hours at each temperature extreme to determine if a failure has occurred. No parts shall be replaced as preventive maintenance during reliability testing. A failure shall be identified as any malfunction that causes a failure in the capacitance value of the dummy converter.

4.6.7 Maintainability. Completion of corrective maintenance tasks within 1 hour (field level only) for expected failures shall demonstrate compliance with the maintainability requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The dummy converters covered by this specification are intended to be used in aircraft when one converter of a multi-converter installation is not installed.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Item identification (see 1.3).

c. Type of dummy converter (see 1.2).

d. When first article is required (see 3.1).

e. When reliability testing is required (see 3.5.3).

f. Data required (see Section 4).

g. Packaging requirements (see 5.1).

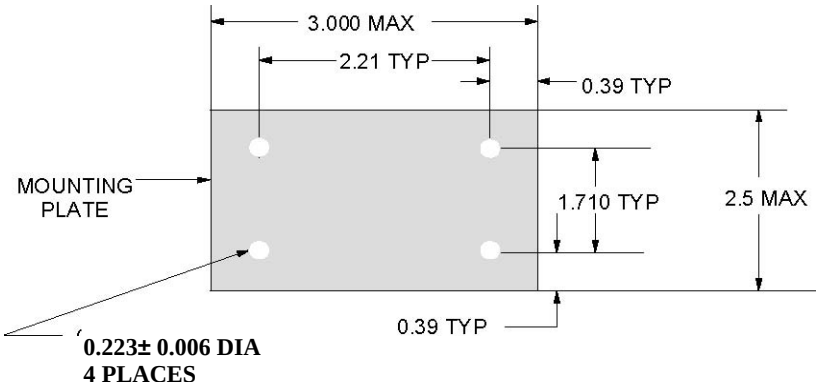
6.3 Cross reference. This specification supersedes MIL-PRF-26392D (USAF). All items that conform to MIL-PRF-26392D are interchangeable with Type II dummy converters, PIN M26392-2.

6.4 Subject term (key word) listing.

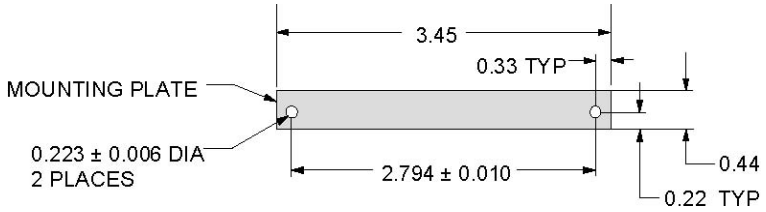
Breathing oxygen
Capacitance
Capacitor
Case
Interchangeability
Maintainability
Reliability

6.5 Changes from previous issue. Only minor changes were made to this document.

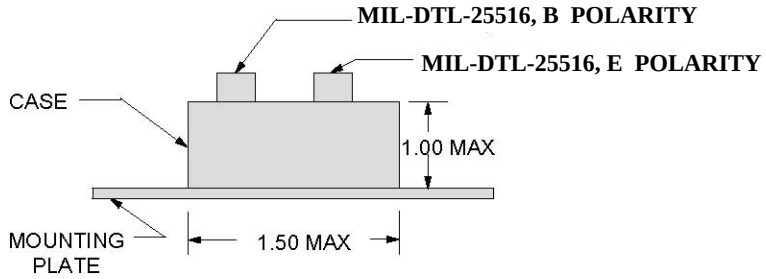
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TOP VIEW TYPE II
(CASE NOT SHOWN)



TOP VIEW TYPE I
(CASE NOT SHOWN)



SIDE VIEW
TYPE I AND TYPE II

Notes:

1. Unless otherwise specified, tolerances are as follows:
   ± 0.030 for .XXX
   ± 0.05 for .XX.
2. Dimensions in inches.

FIGURE 1. Dimensions for case and mounting plate, Type I and Type II

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Preparing Activity:
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